

Daniel Osei

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Professional Summary

Senior machine learning engineer with 8 years of experience taking models from research notebooks into production systems serving millions of requests per day. Specialised in recommendation and ranking systems, with deep experience in distributed training and model serving infrastructure. Has led teams of up to 6 engineers and owns hiring, technical direction and on call practice for the ML platform.

Technical Skills

Languages: Python, Go, SQL, Scala

Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, Hugging Face Transformers

MLOps: MLflow, Weights and Biases, Kubeflow, Feast, ONNX

Infrastructure: Kubernetes, Docker, AWS, Terraform, Airflow, Kafka, Redis

Practices: distributed training, A/B testing, model monitoring, feature stores

Work Experience

Senior Machine Learning Engineer, Halcyon Data, London (2021 - Present)

- Led a team of 5 engineers delivering a real-time recommendation service handling 40M requests per day at a p99 latency of 45ms.
- Cut model training time from 14 hours to 3 hours by moving to distributed PyTorch on Kubernetes with mixed precision.
- Introduced MLflow experiment tracking across 4 teams, replacing an ad hoc mix of spreadsheets and notebook outputs.
- Designed the feature store that now backs 9 production models, cutting the time to ship a new model from 6 weeks to 8 days.
- Ran the migration of 23 models to ONNX runtime, reducing inference cost by 31%.
- Chairs the weekly model review where every production change is signed off.

Machine Learning Engineer, Brightlane, London (2018 - 2021)

- Deployed 11 models to production and owned their monitoring, alerting and retraining schedules.
- Reduced feature pipeline runtime from 9 hours to 2 hours using Airflow and Spark.
- Built an automated drift detection service that paged on distribution shift, catching a broken upstream feed within 20 minutes.
- Introduced shadow deployments so new models could be validated on live traffic before taking any real requests.
- Interviewed 40+ candidates and helped grow the team from 3 to 9 engineers.

Data Scientist, Meridian Retail Group, Manchester (2017 - 2018)

- Built demand forecasting models for 1,400 stores, improving stock allocation accuracy by 9%.
- Automated a pricing analysis that had taken two analysts three days each month.
- Presented quarterly findings to the commercial director and regional managers.

Selected Talks and Publications

Serving Recommendations at Low Latency - PyData London (2023)

Feature Stores in Practice: What We Got Wrong - MLOps Community meetup (2022)

Co-author, Scalable Ranking for Sparse Catalogues, industry workshop paper (2021)

Education

MSc Computer Science, Imperial College London (2017), Distinction

BSc Mathematics, University of Manchester (2015), First Class Honours

Certifications

AWS Certified Machine Learning - Specialty (2022)

Certified Kubernetes Application Developer (2021)